

Exercises

Recall the definition of a σ^{+-} -labeling of a graph G on m edges.

Definition 5.1. The length of a $\mathbb{Z}$ -labeled edge ab modulo n is defined by $\ell_n(ab) = \min\{n - |a - b|, |a - b|\}$. We call an injection $f : \{0, 1, \dots, 2m - 2\} \hookrightarrow V(G)$ a σ^{+-} -labeling of a graph G on m edges if

1. $f(V(G)) = A \sqcup B$ such that $ab \in f(E(G)) \implies a < b$; All neighbors of each $v \in f(V(G))$ are either bigger or smaller than v
2. Every edge is a short edge: $\ell(ab) = |a - b|, \forall ab \in f(E(G))$ and there is exactly one edge of each length in $\{1, \dots, m\}$
3. The edge of length m is a *pendant edge* which means one of its endpoints has only one neighbor; it is incident to a leaf.

$\mathbb{Z}_n$ acts on the set of all $\mathbb{Z}$ -labeled subgraphs $S(K_n)$ of K_n in the following way for each $H \in S(K_n)$ and each $g \in \mathbb{Z}_n$:

$$g(H) := \text{the subgraph induced by } V(H) + g \pmod{n}; \text{ just add } g \text{ to every vertex}$$

Problem 5.2.

1. Find a σ^{+-} -labeling of $S_4 \sqcup P_1$. (lengths should be 1 – 4, labels should be $\{0, 1, \dots, 6\}$)
2. Find its orbit via the $\mathbb{Z}_9$ action.

